

CO3 AT1

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COURSE CODE: CSA0904

COURSE NAME: PROGRAMMING FOR JAVA

1

```
import java.io.*;
```

```
public class TryWithResourcesDemo {
```

```
    public static void main(String[] args) {
```

```
        try (BufferedReader reader =
```

```
            new BufferedReader(new StringReader("Hello from data"))) {
```

```
            String line = reader.readLine();
```

```
            System.out.println(line);
```

```
        } catch (IOException e) {
```

```
            e.printStackTrace();
```

```
        }
```

```
        try (StringWriter writer = new StringWriter()) {
```

```
            writer.write("Hello");
```

```
            System.out.println(writer.toString());
```

```
        } catch (Exception e) {
```

```
            e.printStackTrace();
```

```
}
```

```
class NotCloseable implements AutoCloseable {
```

```
    void write() {
```

```
        System.out.println("Writing");
```

```
    }
```

```
    public void close() {
```

```
        System.out.println("Closed");
```

```
    }
```

```
}
```

```
try (NotCloseable nc = new NotCloseable()) {
```

```
    nc.write();
```

```
} catch (Exception e) {
```

```
    e.printStackTrace();
```

```
}
```

```
try (StringReader fr1 = new StringReader("File A");
```

```
    StringReader fr2 = new StringReader("File B")) {
```

```
    System.out.println("Resources opened successfully");
```

```
} catch (Exception e) {
```

```
    e.printStackTrace();
```

```
}
```

```
}
```

```
}
```



SIMATS Online Programming Lab
JAVA PROGRAMMING

 SELVA PRABHU A K
192421224RA

Your Code

```

1 import java.io.*;
2 public class TryWithResourcesDemo {
3     public static void main(String[] args) {
4         try {BufferedReader reader =
5             new BufferedReader(new StringReader("Hello from data"));
6
7             String line = reader.readLine();
8             System.out.println(line);
9
10        } catch (IOException e) {
11            e.printStackTrace();
12        }
13
14        try {StringWriter writer = new StringWriter(); {
15            writer.write("Hello");
16            System.out.println(writer.toString());
17        }
18        } catch (Exception e) {
19            e.printStackTrace();
20        }
21
22        class NotCloseable implements AutoCloseable {
23
24            void write() {
25                System.out.println("Writing");

```

Input

Output

```

Hello from data
Hello
Writing
Closed
Resources opened successfully

```

2

```
import java.util.concurrent.*;
```

```
public class RunnableCallableDemo {
```

```
    public static void main(String[] args) throws Exception {
```

```
        ExecutorService executor = Executors.newFixedThreadPool(2);
```

```
        Callable<Integer> callableTask = () -> {
```

```
            return 10 / 2;
```

```
        };
```

```
        Future<Integer> future = executor.submit(callableTask);
```

```
        int result = future.get();
```

```
        System.out.println(result);
```

```
        Runnable r = () -> {
```

```
            System.out.println(5);
```

```
};
```

```
executor.submit(r);
```

```
executor.shutdown();
```

```
}
```

```
}
```



The screenshot displays the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the lab name, and a user profile for SELVA PRABHU A K. The main area is divided into three sections: 'Your Code' (a code editor with a dark theme), 'Input' (a text box), and 'Output' (a dark-themed window). The code in the editor is a Java program that demonstrates thread execution using an ExecutorService and Callable. The output window shows the results of the program's execution.

```
1 import java.util.concurrent.*;
2
3 public class RunnableCallableDemo {
4     public static void main(String[] args) throws Exception {
5
6         ExecutorService executor = Executors.newFixedThreadPool(2);
7
8         Callable<Integer> callableTask = () -> {
9             return 10 / 2;
10        };
11
12        Future<Integer> future = executor.submit(callableTask);
13
14        int result = future.get();
15        System.out.println(result);
16
17        Runnable r = () -> {
18            System.out.println(5);
19        };
20
21        executor.submit(r);
22
23        executor.shutdown();
24    }
25 }
```

Input: stdin input

Output: 5, 5

3

```
public class ThreadJoinDemo {
    public static void main(String[] args) {
```

```
        Thread worker = new Thread(() -> {
```

```
            for (int i = 0; i < 5; i++) {
```

```
                System.out.println("Working: " + i);
```

```
            try {
```

```
                Thread.sleep(500);
```

```
        } catch (InterruptedException e) {  
            System.out.println("Worker interrupted");  
            Thread.currentThread().interrupt();  
            break;  
        }  
    }  
});
```

```
worker.start();
```

```
try {  
    Thread.sleep(100);  
} catch (InterruptedException e) {  
    Thread.currentThread().interrupt();  
}
```

```
worker.interrupt();
```

```
try {  
    worker.join(1000);  
} catch (InterruptedException e) {  
    Thread.currentThread().interrupt();  
}
```

```
if (worker.isAlive()) {  
    System.out.println("Still running");  
} else {  
    System.out.println("Worker finished");  
}
```

```

    }
}
}

```

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS ENGINEERING logo, the text "SIMATS Online Programming Lab", and "JAVA PROGRAMMING". A user profile for SELVA PRABHU A K is visible in the top right. The main area is divided into three sections: "Your Code", "Input", and "Output".

Your Code:

```

1 public class ThreadJoinDemo {
2     public static void main(String[] args) {
3
4         Thread worker = new Thread() -> {
5
6             for (int i = 0; i < 5; i++) {
7                 System.out.println("Working: " + i);
8
9                 try {
10                     Thread.sleep(500);
11                 } catch (InterruptedException e) {
12                     System.out.println("Worker interrupted");
13                     Thread.currentThread().interrupt();
14                     break;
15                 }
16             }
17         };
18
19         worker.start();
20
21         try {
22             Thread.sleep(100);
23         } catch (InterruptedException e) {
24             Thread.currentThread().interrupt();
25         }
26

```

Input:

stdin input

Output:

```

Working: 0
Worker interrupted
Worker finished

```

4

```
public class CustomExceptionDemo {
```

```
    public static void main(String[] args) {
```

```
        try {
```

```
            validateAge(-5);
```

```
        } catch (InvalidAgeException e) {
```

```
            System.out.println("Error: " + e.getMessage());
```

```
            e.printStackTrace();
```

```
        }
```

```
    }
```

```
    static void validateAge(int age) throws InvalidAgeException {
```

```
        if (age < 0) {
```

```
        throw new InvalidAgeException("Age cannot be negative");
    }
}
}
```

```
class InvalidAgeException extends Exception {
```

```
    public InvalidAgeException(String msg) {
        super(msg);
    }
```

```
    @Override
```

```
    public String getMessage() {
        return "Custom message";
    }
```

```
    public void printError() {
        System.out.println("Error occurred");
    }
```

```
}
```

```
System.out.println("Error occurred");  
}  
}
```

Your Code

```
1 public class CustomExceptionDemo {  
2  
3     public static void main(String[] args) {  
4  
5         try {  
6             validateAge(-5);  
7         } catch (InvalidAgeException e) {  
8             System.out.println("Error: " + e.getMessage());  
9             e.printStackTrace();  
10        }  
11    }  
12  
13    static void validateAge(int age) throws InvalidAgeException {  
14        if (age < 0) {  
15            throw new InvalidAgeException("Age cannot be negative");  
16        }  
17    }  
18 }  
19  
20 class InvalidAgeException extends Exception {  
21  
22     public InvalidAgeException(String msg) {
```

Input

Output

```
Error: Custom message  
Error occurred
```


